

BreathWatch

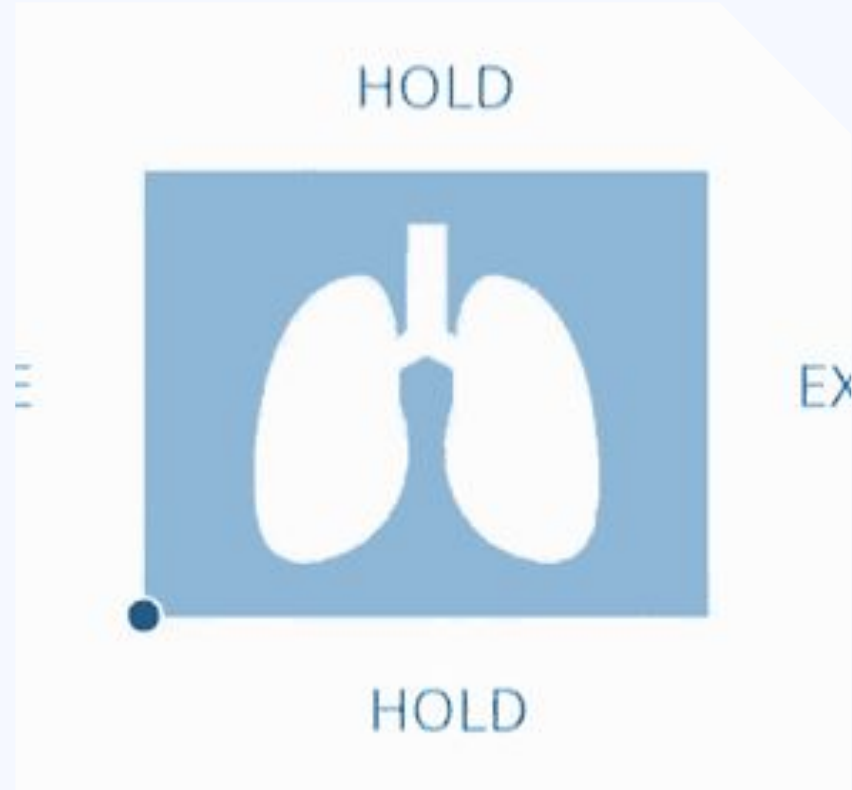
**Automated Respiratory Remote Health
Monitoring Software System**

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Outline

- **Introduction**
- **Design Objectives**
- **Initial Solution**
- **Final Solution**
- **Video Demo**
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Introduction

Around 80% of people experience breathing dysfunction caused by abnormal patterns related to anxiety, anemia, lung, or heart conditions.

- Current Challenges:
 - Hospital equipment is non-portable and expensive to maintain.
 - Access is limited to hospitals, often causing delays for those needing urgent evaluations.
- Proposed Solution:
 - Create a respiratory monitoring system that is easily accessible and quick to use can provide significant benefits by detecting early signs of abnormal patterns and critical conditions.

Design Objectives

The system should:

- Manage both dynamic and static data effectively.
 - Breathing patterns, User information.
- Implement at least three design patterns to ensure robust functionality.
- Allow users to monitor breathing patterns and identify discrepancies.
- Detect and report instances of prolonged breathing pauses.
- Maintain data privacy, ensuring only authorized users can access information.
- Ensure scalability to support future updates and additional features.
 - Such as adding more design patterns.

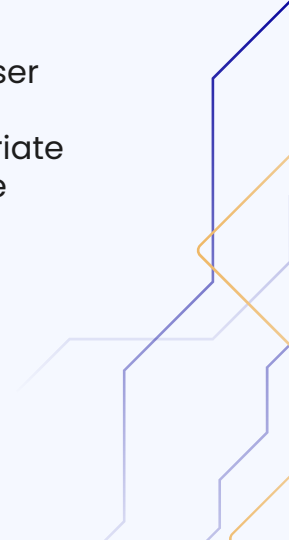


Initial Solution

Planned Patterns:

- Singleton
 - Manage login credentials
 - Grant/Deny access based on user input.
- Decorator
 - Assign user roles (Patient or Clinician)
 - Provide role-specific views and features.
- Observer
 - Monitor breathing data for anomalies
 - Detect pauses or discrepancies in patterns

Reasons Why We Didn't Use:

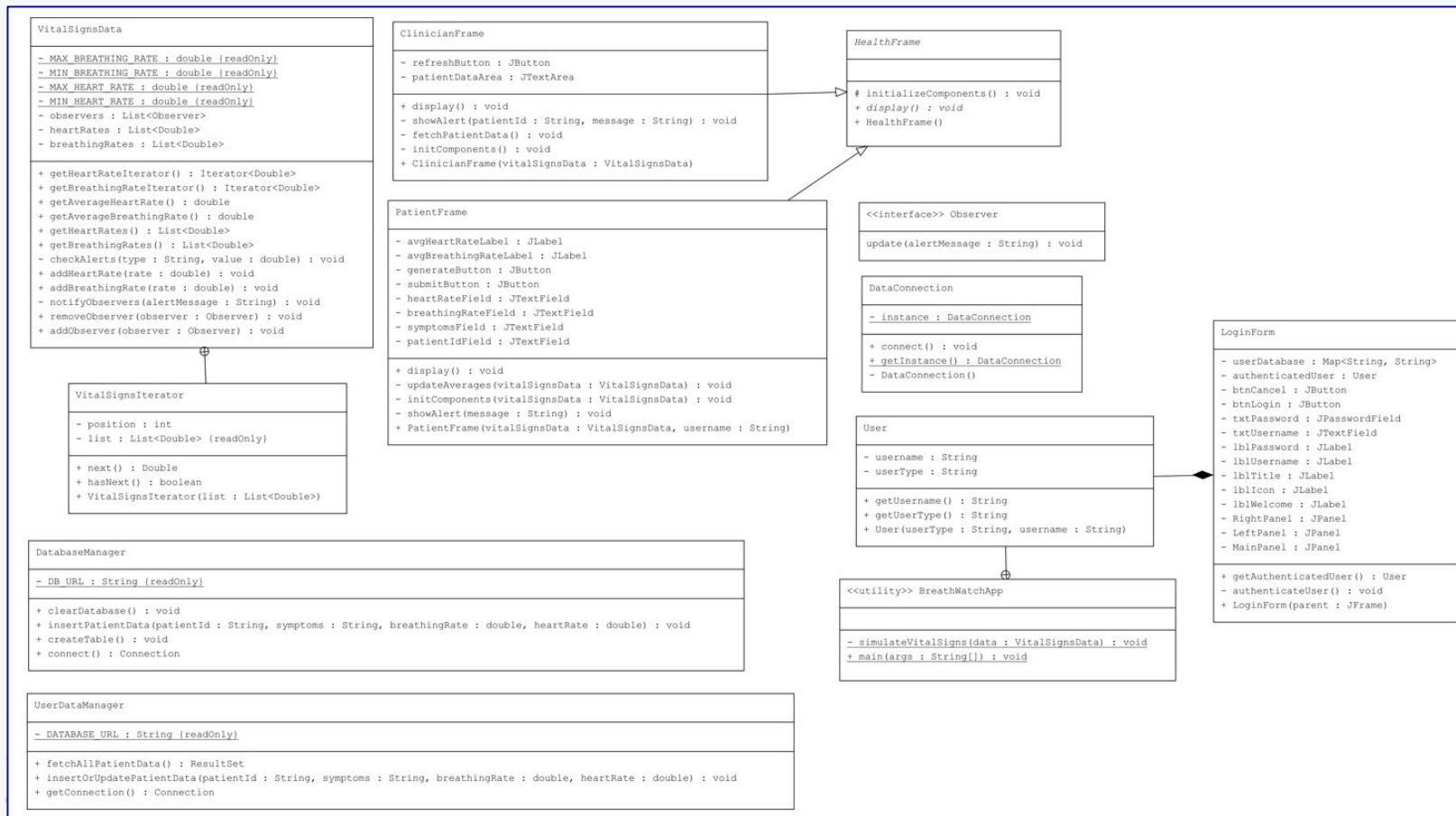
- Singleton Limitation
 - Restricted to a single set of credentials.
 - Prevented dynamic multi-user access.
 - Impact on Decorator Pattern
 - Unable to differentiate user views effectively
 - Failed to display appropriate permissions for each role
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Final Solution

Patterns Used:

- Observer
 - Used to handle alert notifications for changes in vital signs.
 - Notification will pop-up if any there is any irregular breathing or heart rates.
- Iterator
 - Used through the VitalSignsIData file, which allows controlled traversal of the breathingRates and heartRates lists without exposing their internal implementation details
- Singleton
 - Ensures that there is only one instance of DataConnection, which is useful for managing shared resources like a connection to a data source.
 - It provides a single point of control and avoids the overhead of creating multiple connections
- Template
 - The HealthFrame class provides a common structure for initializing frames, with abstract methods and customizable components for subclasses.
 - Subclasses are the PatientFrame and ClinicianFrame.

Final Solution UML



DEMO VIDEO TIME!



Considerations

Environmental:

- Patterns are able to optimize resource utilization and robustness, contributing to reducing carbon emissions.

Societal:

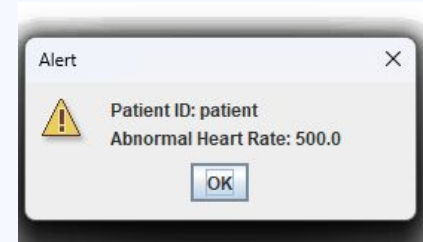
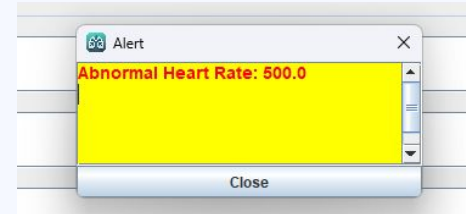
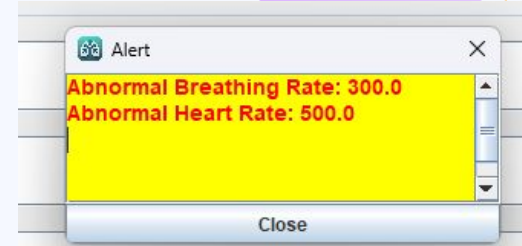
- Designed for easy of use, ensuring accessibility for all users regardless of technical expertise.

Safety:

- Only users with an account can use the Breathwatch system.
- Only allowing authorized users from opening backend data (Clinician)
- Only allowing authorized users from using our system (Patients)

Economic:

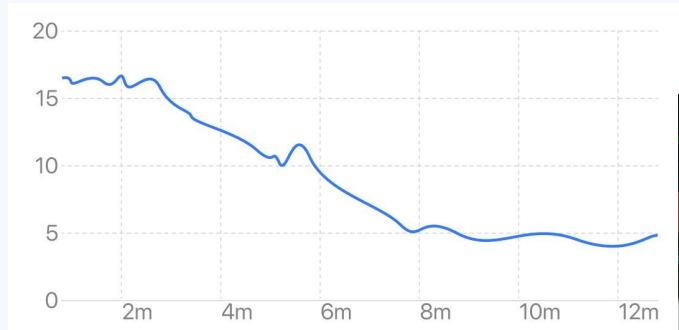
- Made sure that our system is Inexpensive and cost-effective.



Limitations

Some of the limitations that occurred when building the Breathwatch system were:

- Whenever a clinician opens the system, they would get bombarded with notifications of recent patient data, making it somewhat annoying.
- Tired to implement charts and graphs for statistics, but due to time constraints and unable to get the library integrated, were not able to include this.
 - Could show the change in breathing / heart rate for better visuals.



Lifelong Learning

When designing our system, we chose to use a sql database where the patient data can be stored into.

- When a clinician opens the system, they can see all previous entered results from patients.
- The clinician would also be alerted of new submissions



When a patient gets their breathing monitored, if it results in a high number of breaths, the system would alert the patient that their condition should be checked.

- When the clinician opens the system, they can see the result of the abnormal condition.

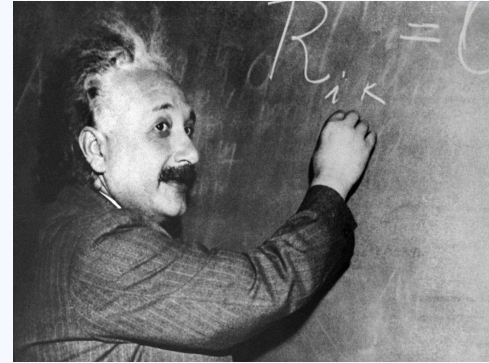


Conclusion / Future Work

With our system, we were able to achieve many of the objectives that we wanted with the Breathwatch monitoring system by using design patterns to help build and model our system around.

For future work, we would like to:

- Incorporate more login credentials to more patients, clinician or other types of users. (User profiles, User history).
- Possible implement a register page and how a database store the data.
- More design patterns to improve / add new features to Breathwatch.
- Implement so it works with mobile devices for easy access and usability.
- Add statistics of the patients on clinician page to view vitals.





Thanks!

Do you have any questions?

